

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – SHORT ANSWER TEST

Course Code:	CSA16	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL1, BL2)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	10% of CIA	Total Marks:	100 (5 Questions × 20 Marks)
CO1 Statement:	Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2		
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Section / Batch:	CSE - AI	Date:	17/7/26

Instructions to Students

- Answer all questions. Each question carries 20 Marks. Total: 100 Marks.
- Clearly show all requirements, process, operations.
- Use appropriate models, schema represented scenario wise
- Justify each step in different dimension specified and measures clearly.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Short Answer Test	Architecture of Data Warehouse, ETL, OLAP	Q1 – Q5	Q1 –20 Q2 –20 Q3 –20 Q4 –20 Q5 –20
GRAND TOTAL:			100 Marks

SECTION A – SHORT ANSWER TEST

1. Suppose that a data warehouse consists of the three dimensions time, doctor, and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit.

- Enumerate three classes of schemas that are popularly used for modelling data warehouses.
 - Draw a schema diagram for the above data warehouse using one of the schema.
 - To obtain the same list, write an SQL query assuming the data is stored in a relational database with the schema fee (day, month, year, doctor, hospital, patient, count, charge).
- (5 Marks)

2. Suppose that a data warehouse for Big University consists of the following four dimensions: student, course, semester, and instructor, and two measures count and avg grade. When at the lowest conceptual level (e.g., for a given student, course, semester, and instructor combination), the avg grade measure stores the actual course grade of the student. At higher conceptual levels, avg grade stores the average grade for the given combination. (5 Marks)

- Draw a snowflake schema diagram for the data warehouse.
- Starting with the base cuboid [student, course, semester, instructor], what specific OLAP operations (e.g., roll-up from semester to year) should one perform in order to list the average grade of CS courses for each Big University student.
- If each dimension has five levels (including all), such as “student < major < status < university < all”, how many cuboids will this cube contain (including the base and apex cuboids)?

star schema:

1. central fact table connected to dimension tables
2. simple and fast Querying.

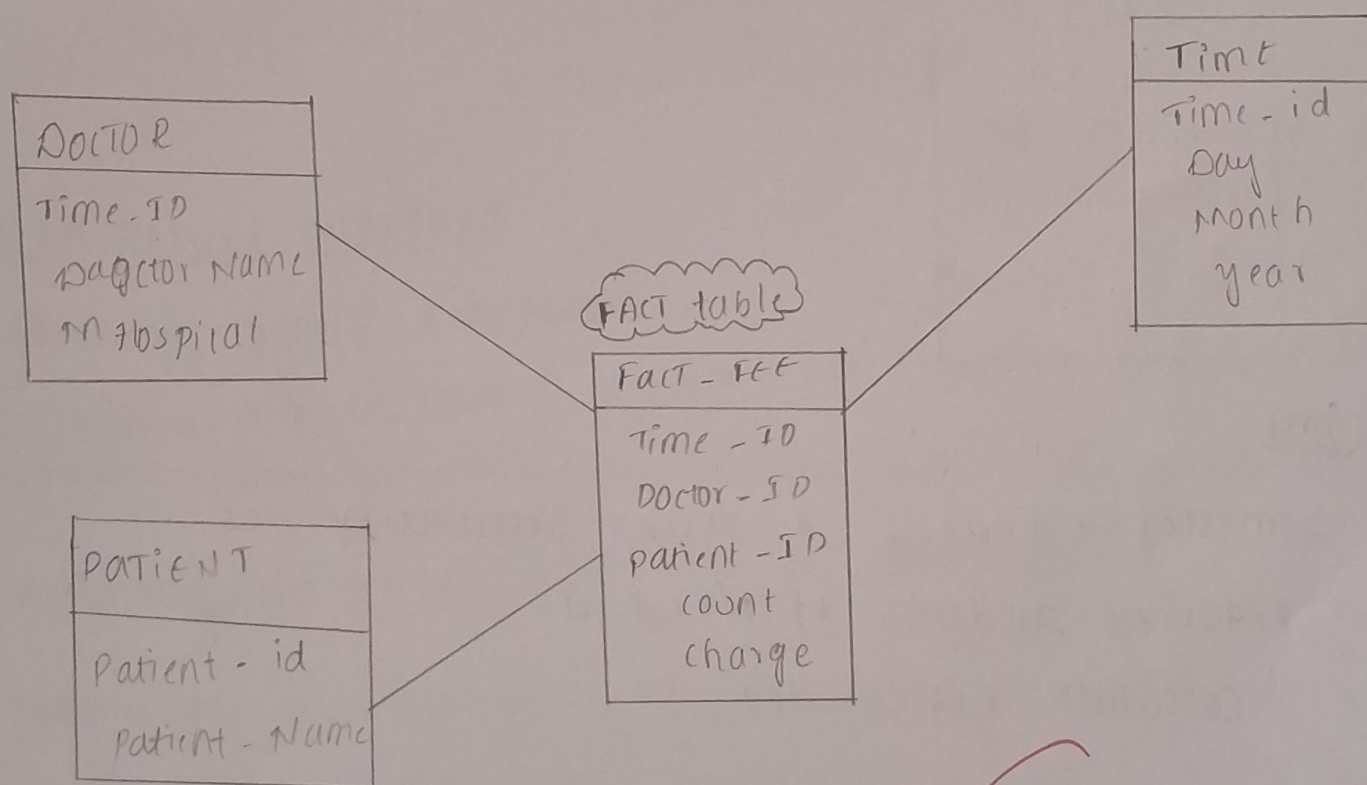
snowflake schema:

1. Dimensions are normalized into multiple related tables
2. saves storage space.

Fact constellation schema:

1. Multiple facts tables share dimension tables
2. Suitable for complex warehouses.

b) star schema diagram



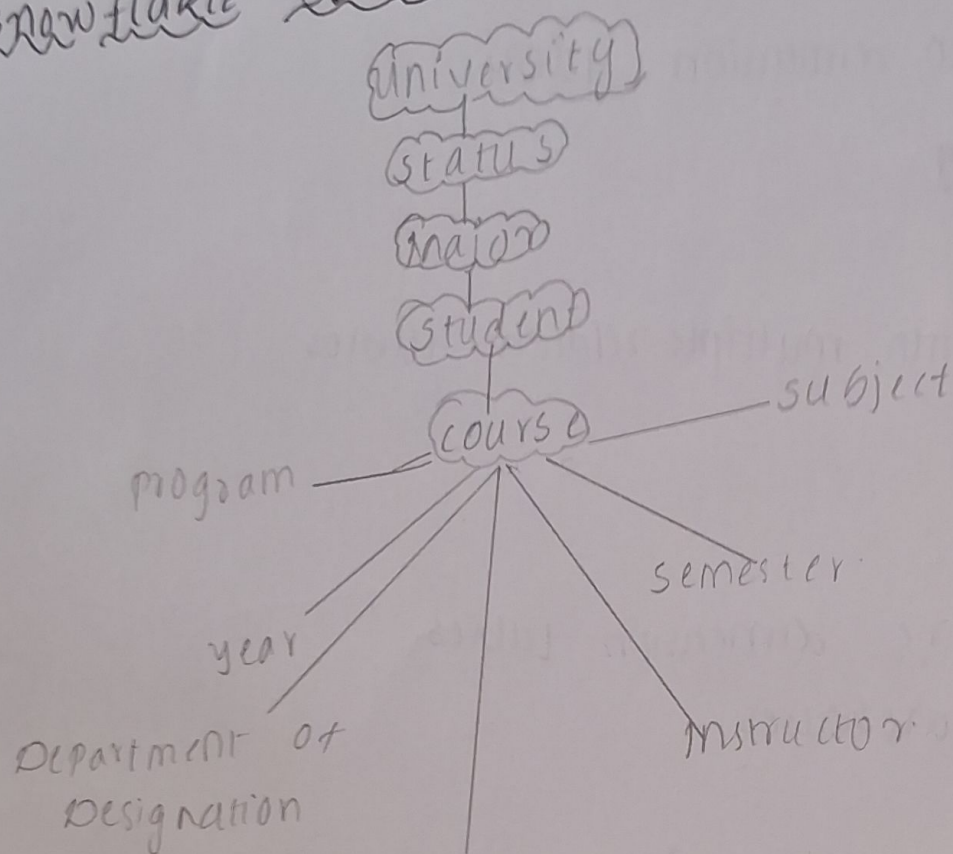
1(c)

sql Query

```

SELECT
Doctor,
Year,
SUM(charge) AS Total-charge,
SUM(count) AS Total-visits
FROM Fee
GROUP BY Doctor, year;
  
```


2(a) snowflake schema



Fact table
student_Id
course_Id
semester_Id
instructor_Id
count
Avg. grade

2(b) OLAP Operations

1. Roll-up: Semester $\rightarrow$ year
2. Slice: Select course = cs
3. Dice: Select required students if needed.
4. Aggregate: Calculate Avg(grade)

2(c) Number of cuboids

$$\text{No. of cuboids} = 2^N \rightarrow \text{number of hierarch levels}$$

N = Number of dimensions.

$$5^4 = 625 \text{ cuboids.}$$

star schema

Fact table

Weather Fact

Probe - id

Time - id

Location - id

Air - pressure

Temperature

Precipitation

Dimension Tables

Time

house

Day

month

Year

Location

Ocean/land

city

state

Region

Probe

Probe - id

Probe - type

schema

Time

Location - weather-fact - Probe

OLAP queries supposed

1. Average yearly temperature

2. Monthly rainfall

3. Highest air pressure by region

4. seasonal weather patterns.

4(a)

Maximum number of cells in a Base cuboid = p^n

4(b)

Minimum number of cells in Base cuboid if all tuples fall into one cell = 1

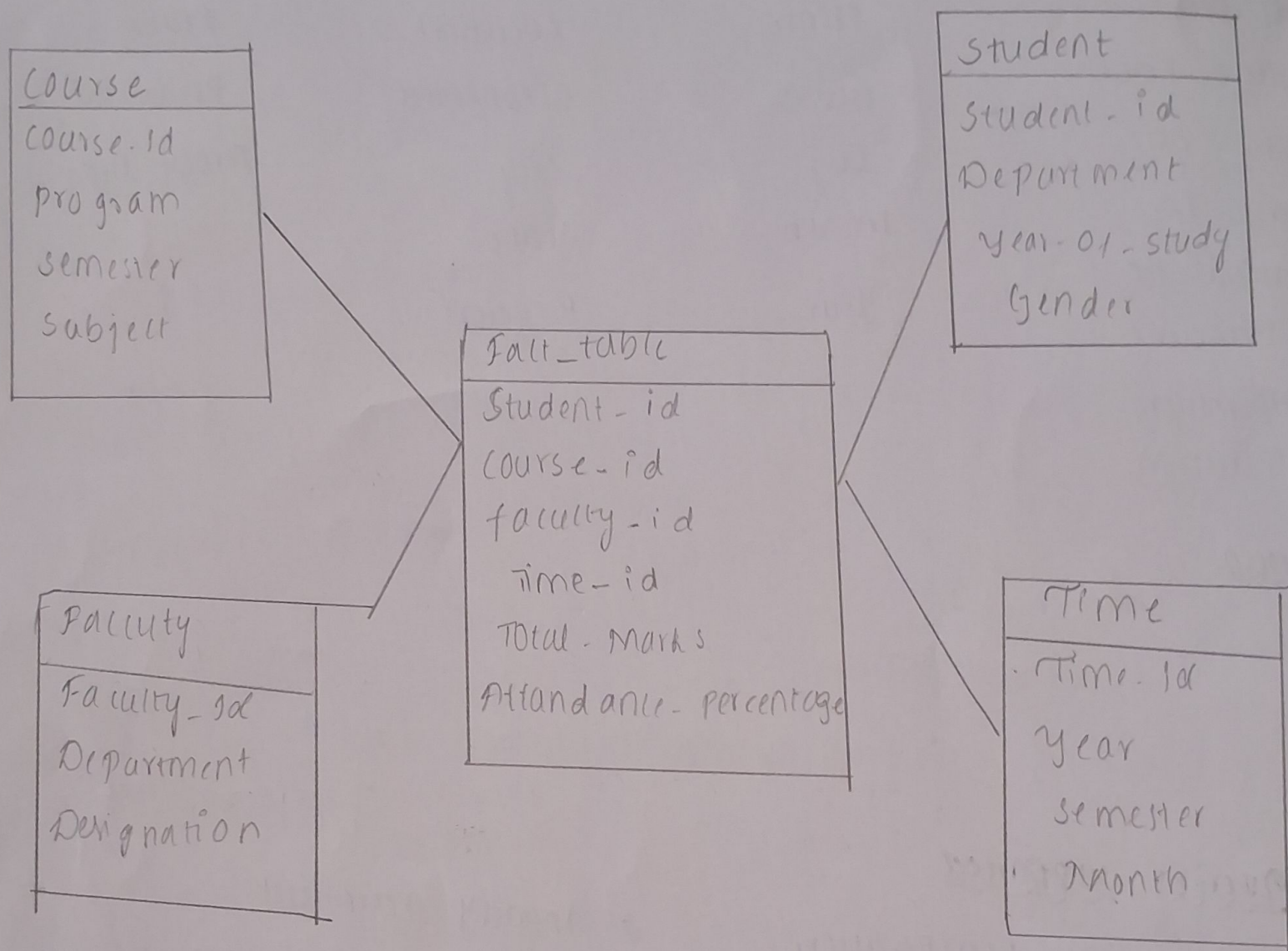
4(c)

Maximum number of cells in entire cube : $(p+1)^n$

4(d)

Minimum number of cells in entire cube : 2^n

5(a) Star schema



5(b) OLAP operations

1. Roll up: Month $\rightarrow$ Semester $\rightarrow$ Year
2. Roll up: Student: Student $\rightarrow$ department
3. Aggregate: Avg (Total marks)

5(c) Number of cuboids

Students = 4

Course = 3

Faculty = 3

Time = 5

$$4 \times 3 \times 3 \times 5 = 180 \text{ cuboids.}$$